

2019-2020 Spring
MAT124 Resit
01/07/2020

1. The temperature T at the point (x, y, z) in a region of space is given by the formula $T = 100 - xy - xz - yz$. Find the lowest temperature on the plane $x + y + z = 10$.
2. Show that if $z = f(r, \theta)$, where r and θ are defined as functions of x and y by the equations $x = r \cos \theta$ and $y = r \sin \theta$, then the equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$ becomes

$$\frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} + \frac{1}{r} \frac{\partial z}{\partial r} = 0.$$

3. Evaluate the integral $\int_0^1 \int_{\sqrt{y}}^1 \sqrt{1 - x^3} \, dx \, dy$.

4. Evaluate

$$\int_2^4 \int_2^y dx \, dy + \int_4^8 \int_2^{16/y} dx \, dy$$

by reversing the order of integration.

5. Use a double integral to find the area inside the circle $r = \cos \theta$ and outside the cardioid $r = 1 - \cos \theta$.
6. Use polar coordinates to evaluate the double integral

$$\iint_D \sin(x^2 + y^2) \, dA,$$

where D is the region bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$ and the lines $y = 0$, $x = \sqrt{3}y$.

7. Using cylindrical coordinates, evaluate

$$\iiint_D \frac{dV}{x^2 + y^2 + z^2},$$

where D is the solid region bounded below by the paraboloid $2z = x^2 + y^2$ and above by the sphere $x^2 + y^2 + z^2 = 8$.

8. Using spherical coordinates, evaluate the triple integral

$$\iiint_D \sqrt{x^2 + y^2 + z^2} \, dV,$$

where D is the portion of the solid sphere $x^2 + y^2 + z^2 \leq 1$ that lies in the first octant.

1. Let $g(x, y, z) = x + y + z - 10$ and then, solve the system of equations below using the method of Lagrange multipliers.

$$\left. \begin{array}{l} \nabla T = \lambda \nabla g \\ g(x, y, z) = 0 \end{array} \right\} \quad \nabla T = \langle -y - z, -x - z, -x - y \rangle = \lambda \langle 1, 1, 1 \rangle = \lambda \nabla g$$

$$T_x + T_y + T_z = (-y - z) + (-x - z) + (-x - y) = -2x - 2y - 2z$$

$$= \lambda + \lambda + \lambda = 3\lambda \implies x + y + z = \frac{-3\lambda}{2}$$

Use the constraint to find the value of λ .

$$g(x, y, z) = 0 \implies \frac{-3\lambda}{2} - 10 = 0 \implies \lambda = -\frac{20}{3}$$

So far, we have the equations below. Solve the system of equations and find the values of x, y, z one by one.

$$-y - z = -\frac{20}{3} \tag{1}$$

$$-x - z = -\frac{20}{3} \tag{2}$$

$$-x - y = -\frac{20}{3} \tag{3}$$

Solve for x and z .

$$(1) \ \& \ (2) \implies x - y = 0 \tag{4}$$

$$(3) \ \& \ (4) \implies y = \frac{10}{3}$$

$$\therefore \quad z = \frac{10}{3}, \quad x = \frac{10}{3}.$$

We now have all the values. Substitute in $T(x, y, z)$ to find the minimum value of the temperature.

$$T_{\min} = T\left(\frac{10}{3}, \frac{10}{3}, \frac{10}{3}\right) = 100 - \left(\frac{10}{3}\right)^2 - \left(\frac{10}{3}\right)^2 - \left(\frac{10}{3}\right)^2 = \boxed{\frac{200}{3}}$$

2. We have $x = r \cos \theta$ and $y = r \sin \theta$.

$$x^2 = r^2 \cos^2 \theta, \quad y^2 = r^2 \sin^2 \theta \implies x^2 + y^2 = r^2 (\cos^2 \theta + \sin^2 \theta) = r^2, \quad r = \sqrt{x^2 + y^2}$$

$$\frac{y}{x} = \frac{r \sin \theta}{r \cos \theta} = \tan \theta \implies \theta = \tan^{-1} \frac{y}{x}$$

Compute the first-order partial derivatives.

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial z}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial x}, & \frac{\partial z}{\partial y} &= \frac{\partial z}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial z}{\partial \theta} \cdot \frac{\partial \theta}{\partial y} \\ \frac{\partial r}{\partial x} &= \frac{x}{\sqrt{x^2 + y^2}} = \cos \theta, & \frac{\partial r}{\partial y} &= \frac{y}{\sqrt{x^2 + y^2}} = \sin \theta \\ \frac{\partial \theta}{\partial x} &= \frac{1}{1 + \frac{y^2}{x^2}} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2 + y^2} = \frac{-\sin \theta}{r}, & \frac{\partial \theta}{\partial y} &= \frac{1}{\frac{y^2}{x^2}} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2} = \frac{\cos \theta}{r}\end{aligned}$$

Rewrite $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial r} \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \frac{-y}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial r} \cdot \frac{y}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \frac{x}{x^2 + y^2}$$

Compute the second-order partial derivatives.

$$\begin{aligned}\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial r} \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \frac{-y}{x^2 + y^2} \right) \\ \frac{\partial^2 z}{\partial x^2} &= \left[\left(\frac{\partial^2 z}{\partial r^2} \cdot \frac{\partial r}{\partial x} + \frac{\partial^2 z}{\partial \theta \partial r} \cdot \frac{\partial \theta}{\partial x} \right) \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial r} \cdot \frac{1 \cdot \sqrt{x^2 + y^2} - x \cdot \frac{x}{\sqrt{x^2 + y^2}}}{\left(\sqrt{x^2 + y^2}\right)^2} \right] \\ &\quad + \left[\left(\frac{\partial^2 z}{\partial \theta^2} \cdot \frac{\partial \theta}{\partial x} + \frac{\partial^2 z}{\partial r \partial \theta} \cdot \frac{\partial r}{\partial x} \right) \cdot \frac{-y}{x^2 + y^2} + \frac{\partial z}{\partial \theta} \cdot \frac{y}{(x^2 + y^2)^2} \cdot 2x \right] \\ \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) &= \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial r} \cdot \frac{y}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial \theta} \cdot \frac{x}{x^2 + y^2} \right) \\ \frac{\partial^2 z}{\partial y^2} &= \left[\left(\frac{\partial^2 z}{\partial r^2} \cdot \frac{\partial r}{\partial y} + \frac{\partial^2 z}{\partial \theta \partial r} \cdot \frac{\partial \theta}{\partial y} \right) \cdot \frac{y}{\sqrt{x^2 + y^2}} + \frac{\partial z}{\partial r} \cdot \frac{1 \cdot \sqrt{x^2 + y^2} - y \cdot \frac{y}{\sqrt{x^2 + y^2}}}{\left(\sqrt{x^2 + y^2}\right)^2} \right] \\ &\quad + \left[\left(\frac{\partial^2 z}{\partial \theta^2} \cdot \frac{\partial \theta}{\partial y} + \frac{\partial^2 z}{\partial r \partial \theta} \cdot \frac{\partial r}{\partial y} \right) \cdot \frac{x}{x^2 + y^2} + \frac{\partial z}{\partial \theta} \cdot \frac{-x}{(x^2 + y^2)^2} \cdot 2y \right]\end{aligned}$$

Add the second-order partial derivatives and set to 0. The last terms eliminate each

other. Write x and y in terms of r and θ .

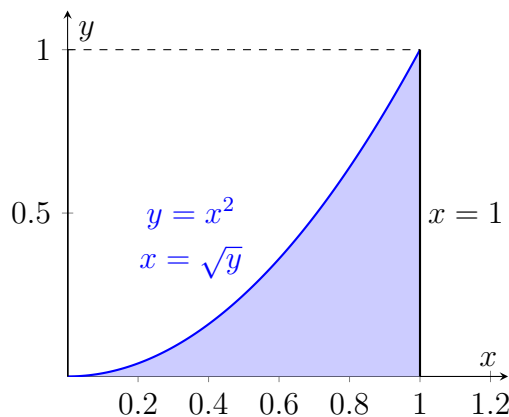
$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} &= \left[\left(\frac{\partial^2 z}{\partial r^2} \cdot \cos \theta + \frac{\partial^2 z}{\partial \theta \partial r} \cdot \frac{-\sin \theta}{r} \right) \cdot \cos \theta + \frac{\partial z}{\partial r} \cdot \frac{\sin^2 \theta}{r} \right] \\ &\quad + \left[\left(\frac{\partial^2 z}{\partial \theta^2} \cdot \frac{-\sin \theta}{r} + \frac{\partial^2 z}{\partial r \partial \theta} \cdot \cos \theta \right) \cdot \frac{-\sin \theta}{r} \right] \\ &\quad + \left[\left(\frac{\partial^2 z}{\partial r^2} \cdot \sin \theta + \frac{\partial^2 z}{\partial \theta \partial r} \cdot \frac{\cos \theta}{r} \right) \cdot \sin \theta + \frac{\partial z}{\partial r} \cdot \frac{\cos^2 \theta}{r} \right] \\ &\quad + \left[\left(\frac{\partial^2 z}{\partial \theta^2} \cdot \frac{\cos \theta}{r} + \frac{\partial^2 z}{\partial r \partial \theta} \cdot \sin \theta \right) \cdot \frac{\cos \theta}{r} \right] = 0 \end{aligned}$$

Inspect the terms that add up to 0. Recall the identity $\sin^2 \theta + \cos^2 \theta = 1$, then the equation reduces to

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} &= \frac{\partial^2 z}{\partial r^2} \cdot (\cos^2 \theta + \sin^2 \theta) + \frac{\partial z}{\partial r} \cdot \frac{\sin^2 \theta + \cos^2 \theta}{r} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} \cdot (\sin^2 \theta + \cos^2 \theta) \\ &= \boxed{\frac{\partial^2 z}{\partial r^2} + \frac{1}{r} \frac{\partial z}{\partial r} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2} = 0}, \end{aligned}$$

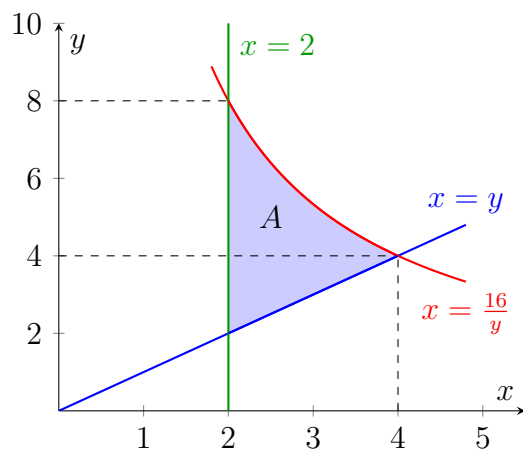
which we set out to demonstrate.

3. Change the order of integration using the graph below and then evaluate the integral.



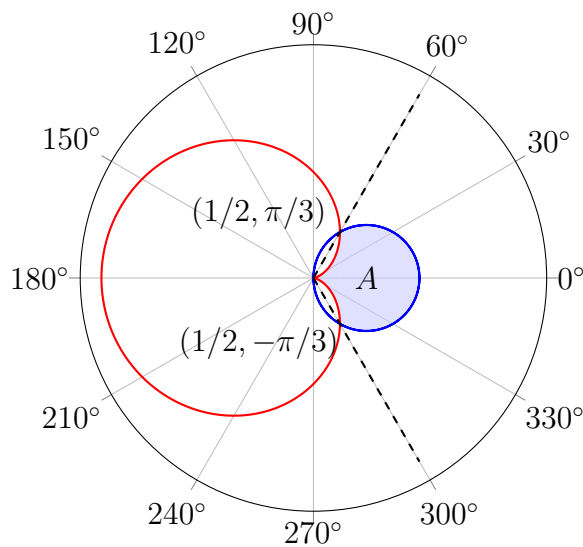
$$\begin{aligned} \int_0^1 \int_{\sqrt{y}}^1 \sqrt{1-x^3} \, dx \, dy &= \int_0^1 \int_0^{x^2} \sqrt{1-x^3} \, dy \, dx = \int_0^1 x^2 \sqrt{1-x^3} \, dx \left[\begin{array}{l} u=1-x^3 \\ du=-3x^2 \, dx \end{array} \right] \\ &= \int \frac{\sqrt{u}}{-3} \, du = -\frac{2}{9} u^{3/2} + c = -\frac{2}{9} (1-x^3)^{3/2} \bigg|_0^1 = \boxed{\frac{2}{9}} \end{aligned}$$

4.



$$\begin{aligned}
 A &= \int_2^4 \int_2^y dx dy + \int_4^8 \int_2^{16/y} dx dy \\
 &= \int_2^4 \int_x^{\frac{16}{x}} dy dx = \int_2^4 \left(\frac{16}{x} - x \right) dx \\
 &= \left[16 \ln |x| - \frac{x^2}{2} \right]_2^4 \\
 &= \left[\left(16 \ln 4 - \frac{4^2}{2} \right) - \left(16 \ln 2 - \frac{2^2}{2} \right) \right] \\
 &= \boxed{16 \ln 2 - 6}
 \end{aligned}$$

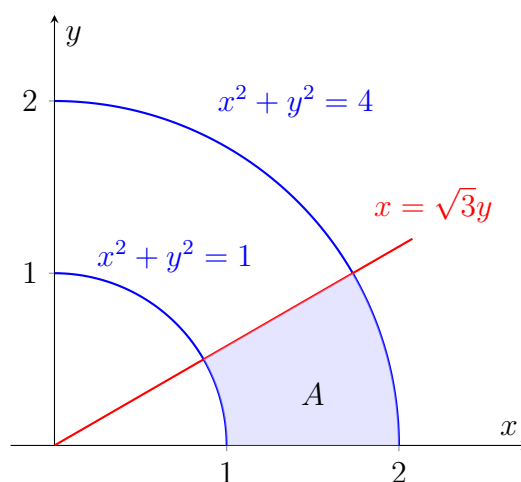
5.



$$\begin{aligned}
 A &= \int_{-\pi/3}^{\pi/3} \int_{1-\cos \theta}^{\cos \theta} r dr d\theta \\
 &= \frac{1}{2} \int_{-\pi/3}^{\pi/3} [\cos^2 \theta - (1 - \cos \theta)^2] d\theta \\
 &= \frac{1}{2} \int_{-\pi/3}^{\pi/3} (2 \cos \theta - 1) d\theta \\
 &= \frac{1}{2} \left[2 \sin \theta - \theta \right]_{-\pi/3}^{\pi/3} \dots
 \end{aligned}$$

$$\dots \quad A = \frac{1}{2} \left[\left(2 \sin \frac{\pi}{3} - \frac{\pi}{3} \right) - \left(2 \sin \left(-\frac{\pi}{3} \right) + \frac{\pi}{3} \right) \right] = \boxed{\sqrt{3} - \frac{\pi}{3}}$$

6.



Use the following transformation to switch to polar coordinates.

$$\left. \begin{array}{l} x = r \cos \theta \\ y = r \sin \theta \\ x^2 + y^2 = r^2 \\ \theta = \tan^{-1} \frac{y}{x} \\ dA = r dr d\theta \end{array} \right\} \Rightarrow \begin{array}{l} x^2 + y^2 = 1 \Rightarrow r^2 = 1 \Rightarrow r = 1 \\ x^2 + y^2 = 4 \Rightarrow r^2 = 4 \Rightarrow r = 2 \\ y = 0 \Rightarrow \theta = 0 \\ x = \sqrt{3}y \Rightarrow \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{1}{\sqrt{3}} \Rightarrow \theta = \frac{\pi}{6} \\ \therefore 1 \leq r \leq 2, \quad 0 \leq \theta \leq \frac{\pi}{6} \end{array}$$

$$\begin{aligned} \int_0^{\pi/6} \int_1^2 \sin(r^2) r dr d\theta &= \int_0^{\pi/6} \left[-\frac{1}{2} \cos r^2 \right]_1^2 d\theta = -\frac{1}{2} \int_0^{\pi/6} (\cos 4 - \cos 1) d\theta \\ &= \frac{1}{2} (\cos 1 - \cos 4) \cdot \theta \Big|_0^{\pi/6} = \boxed{\frac{\pi}{12} (\cos 1 - \cos 4)} \end{aligned}$$

7. For cylindrical coordinates, we have

$$\left. \begin{array}{l} z = z \\ r^2 = x^2 + y^2 \\ dV = r dz dr d\theta \end{array} \right\} \Rightarrow \begin{array}{l} 2z = x^2 + y^2 \Rightarrow z = \frac{r^2}{2} \\ x^2 + y^2 + z^2 = 8 \Rightarrow z = \sqrt{8 - r^2} \end{array}$$

Find where the surfaces $2z = x^2 + y^2$ and $x^2 + y^2 + z^2 = 8$ intersect to determine the limits of r .

$$\begin{aligned} x^2 + y^2 + z^2 = 8 &\Rightarrow 2z + z^2 = 8 \Rightarrow (z + 4)(z - 2) = 0 \Rightarrow z = 2 \\ &\Rightarrow 4 = x^2 + y^2 = r^2 \Rightarrow r = 2 \end{aligned}$$

The lower limit of r is apparently 0. The region in the xy -plane is circular if we project the domain. Therefore, $0 \leq \theta \leq 2\pi$. Now, set up the triple integral in polar coordinates.

$$\begin{aligned} I &= \iiint_D \frac{dV}{x^2 + y^2 + z^2} = \int_0^{2\pi} \int_0^2 \int_{r^2/2}^{\sqrt{8-r^2}} \frac{r}{r^2 + z^2} dz dr d\theta \\ &= \int_0^{2\pi} \int_0^2 \int_{r^2/2}^{\sqrt{8-r^2}} \frac{1}{1 + \left(\frac{z}{r}\right)^2} \cdot \frac{1}{r} dz dr d\theta = \int_0^{2\pi} \int_0^2 \left[\arctan \left(\frac{z}{r} \right) \right]_{r^2/2}^{z=\sqrt{8-r^2}} dr d\theta \\ &= \int_0^{2\pi} \int_0^2 \arctan \left(\frac{\sqrt{8-r^2}}{r} \right) dr d\theta - \int_0^{2\pi} \int_0^2 \arctan \left(\frac{r}{2} \right) dr d\theta \end{aligned}$$

θ is independent of r . Therefore, we can write the following.

$$I = 2\pi \int_0^2 \arctan\left(\frac{\sqrt{8-r^2}}{r}\right) dr - 2\pi \int_0^2 \arctan\left(\frac{r}{2}\right) dr \quad (5)$$

Now, use integration by parts for the left-hand integral in (5). Apply the chain rule and the quotient rule rigorously.

$$\begin{aligned} u = \arctan\left(\frac{\sqrt{8-r^2}}{r}\right) &\implies du = \frac{1}{1 + \left(\frac{\sqrt{8-r^2}}{r}\right)^2} \cdot \frac{\frac{1}{2\sqrt{8-r^2}} \cdot (-2r) \cdot r - \sqrt{8-r^2} \cdot 1}{r^2} dr \\ dv = dr &\implies v = r \end{aligned}$$

Notice that we have an improper integral, where we need to use limits.

$$\begin{aligned} \int_0^2 \arctan\left(\frac{\sqrt{8-r^2}}{r}\right) dr &= \lim_{T \rightarrow 0^+} \left[r \cdot \arctan\left(\frac{\sqrt{8-r^2}}{r}\right) \Big|_T^2 - \int_T^2 r \cdot \frac{-1}{\sqrt{8-r^2}} dr \right] \\ &= \lim_{T \rightarrow 0^+} \left[r \cdot \arctan\left(\frac{\sqrt{8-r^2}}{r}\right) - \sqrt{8-r^2} \right]_T^2 \end{aligned} \quad (6)$$

Compute the other integral in (5) using integration by parts.

$$\begin{aligned} u = \arctan\left(\frac{r}{2}\right) &\implies du = \frac{1}{1 + \left(\frac{r}{2}\right)^2} \cdot \frac{1}{2} dr \\ dv = dr &\implies v = r \end{aligned}$$

$$\int_0^2 \arctan\left(\frac{r}{2}\right) dr = r \arctan\left(\frac{r}{2}\right) \Big|_0^2 - \int_0^2 \frac{2r}{4+r^2} dr = r \arctan \frac{r}{2} - \ln |4+r^2| \Big|_0^2 \quad (7)$$

Rewrite (5) using (6) and (7).

$$\begin{aligned} I &= 2\pi \lim_{T \rightarrow 0^+} \left[r \cdot \arctan\left(\frac{\sqrt{8-r^2}}{r}\right) - \sqrt{8-r^2} \right]_T^2 - 2\pi \left[r \cdot \arctan\left(\frac{r}{2}\right) - \ln |4+r^2| \right]_0^2 \\ &= 2\pi (2 \cdot \arctan 1 - 2) - 2\pi \lim_{T \rightarrow 0^+} \left[T \cdot \arctan \frac{\sqrt{8-T^2}}{T} - \sqrt{8-T^2} \right] \\ &\quad - 2\pi [(2 \cdot \arctan 1 - \ln 8) - (0 - \ln 4)] \\ &= \pi \left(\ln 4 - 4 + 2 \lim_{T \rightarrow 0^+} \left(\sqrt{8-T^2} \right) \right) - 2\pi \lim_{T \rightarrow 0^+} \left(T \cdot \arctan \frac{\sqrt{8-T^2}}{T} \right) \end{aligned} \quad (8)$$

We need to evaluate the limit on the right side in (8) using the squeeze theorem.

$$\begin{aligned}
 -\frac{\pi}{2} &\leq \arctan\left(\frac{\sqrt{8-T^2}}{T}\right) \leq \frac{\pi}{2} \\
 -\frac{T \cdot \pi}{2} &\leq T \cdot \arctan\left(\frac{\sqrt{8-T^2}}{T}\right) \leq \frac{T \cdot \pi}{2} \\
 \lim_{T \rightarrow 0^+} \frac{-T \cdot \pi}{2} &= \lim_{T \rightarrow 0^+} \frac{T \cdot \pi}{2} = 0 \implies \lim_{T \rightarrow 0^+} \left(T \cdot \arctan \frac{\sqrt{8-T^2}}{T}\right) = 0
 \end{aligned}$$

The limit on the left side in (8) is simply equal to $2\sqrt{2}$. The value of the integral is then

$$I = \pi \left(\ln 4 - 4 + 4\sqrt{2} \right)$$

8. For spherical coordinates, we have

$$\left. \begin{aligned} z &= \rho \cos \phi \\ r &= \rho \sin \phi \\ x^2 + y^2 + z^2 &= \rho^2 \\ dV &= \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \end{aligned} \right\} \implies \begin{aligned} x^2 + y^2 + z^2 &\leq 1 \implies \rho^2 \leq 1 \implies 0 \leq \rho \leq 1 \\ \sqrt{x^2 + y^2 + z^2} &= 1 \implies \sqrt{\rho^2} = 1 \implies \rho = 1 \\ \therefore 0 &\leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq \frac{\pi}{2} \end{aligned}$$

Set up the integral and then evaluate.

$$\begin{aligned}
 I &= \iiint_D \sqrt{x^2 + y^2 + z^2} \, dV = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \rho \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\
 &= \int_0^{\pi/2} \int_0^{\pi/2} \left[\frac{\rho^4}{4} \right]_{\rho=0}^{\rho=1} \sin \phi \, d\phi \, d\theta = \frac{1}{4} \int_0^{\pi/2} \int_0^{\pi/2} \sin \phi \, d\phi \, d\theta \\
 &= \frac{1}{4} \int_0^{\pi/2} \left[-\cos \phi \right]_0^{\pi/2} d\theta = \frac{1}{4} \int_0^{\pi/2} d\theta = \boxed{\frac{\pi}{8}}
 \end{aligned}$$